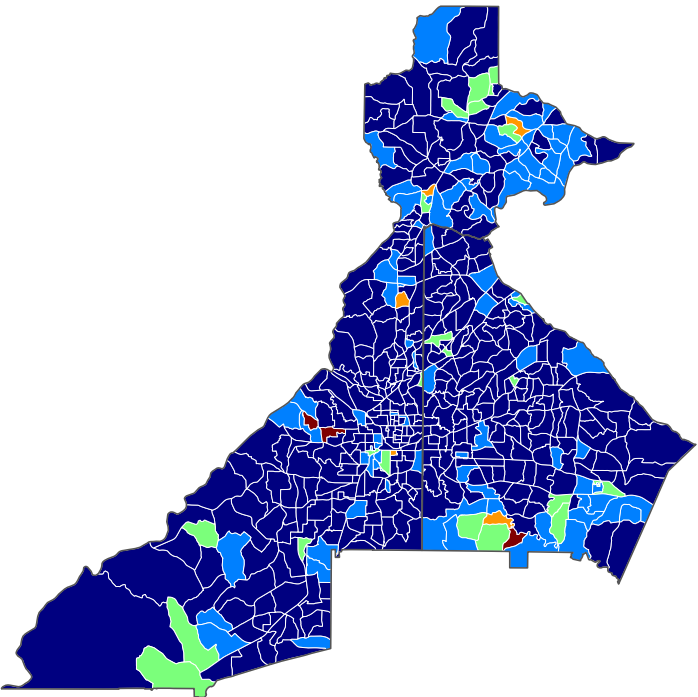


Same data, same tracts — why deterministic validation matters

Atlanta metro · 530 Fulton + DeKalb census tracts · real TIGER geometry · tree-canopy-loss (seeded SAR variable)

UNGATED

naïve LLM output

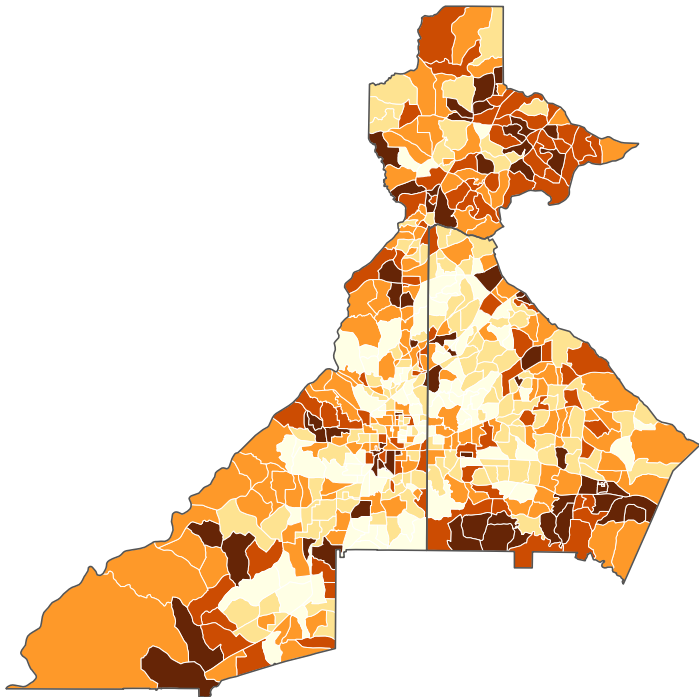


✗ Equal-interval breaks: 414/530 tracts (78%) collapse into one class - spatial variation is hidden. [Gate 2]

✗ Rainbow (jet) ramp: non-monotonic lightness, unreadable under red-green CVD. [Gate 5, specified]

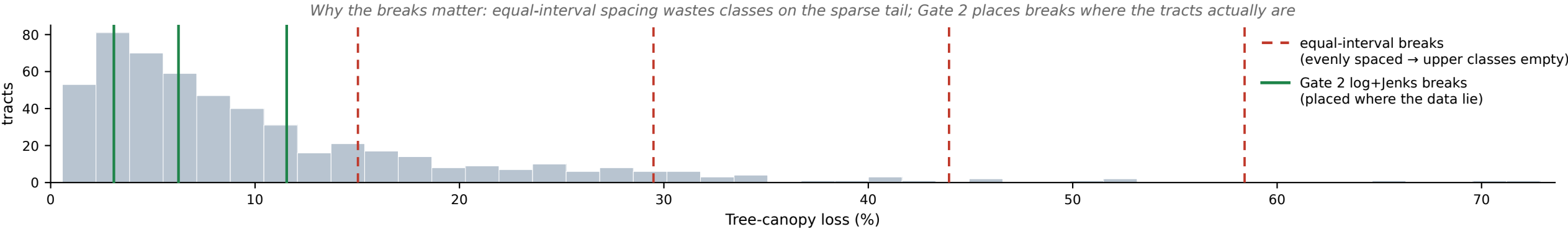
GATED

AutoCarto-Agent output



✓ Gate 2 → log-transform + Jenks: 5 balanced classes [98/134/144/91/63] - spatial structure revealed. [Gate 2 ✓]

✓ ColorBrewer YlOrBr sequential - perceptually ordered, colour-blind safe. [Gate 5, specified]



Gate 2 (classification) is implemented and computed on this figure. Gate 5 (colour) is specified in the roadmap - ramps shown to illustrate the failure it targets.
GVF is deliberately not the score here: equal-interval attains a comparable GVF (0.87 vs 0.83) yet hides the pattern, so class balance is the honest metric.